

PDF & File Formats

Lesson 33 — PDF Behaviour and Format Identification

JKSSB Computer Awareness

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What Is a File Format?

- ▶ Every file is stored as a series of bits; the **file format** is the rule that says how those bits are arranged.
- ▶ The format tells the software how to read the file's contents — text, image, spreadsheet, or slide.
- ▶ The **extension** is the suffix at the end of the file name (for example .pdf or .docx).
- ▶ The extension is only a label; the format is the real structure of the data inside.

The one idea

The extension names the file; the format defines what the file actually is.

Extension vs Format

File name	report.docx
File name (base)	report
Extension	docx (written .docx)
Format	DOCX — an Office Open XML word-processing document

Trap alert (TRAP-34)

.png is an **extension**; PNG is the **format**. Never say that the format *is* the extension.

Common Document Formats

Extension	Format	Typical use
.docx	Word document	Text, formatting, images (MS Word)
.xlsx	Excel workbook	Worksheets, formulas, charts (MS Excel)
.pptx	PowerPoint presentation	Slides and slide shows (PowerPoint)
.csv	Comma-Separated Values	Plain-text rows separated by commas
.txt	Plain text	Unformatted characters only

DOCX, XLSX, and PPTX are the Office Open XML formats. CSV and TXT are plain-text formats.

Common Image Extensions

- ▶ **PNG** — Portable Network Graphics; lossless, supports transparency.
- ▶ **JPG / JPEG** — compressed photographs; lossy (some quality is lost).
- ▶ **GIF** — Graphics Interchange Format; simple images and short animations.
- ▶ **BMP** — Bitmap; a raw image, larger than PNG/JPEG.
- ▶ **TIFF** — Tagged Image File Format; high-quality scanning and print.
- ▶ **SVG** — Scalable Vector Graphics; drawn with lines and shapes, scales without blur.

PDF: The Portable Document Format

- ▶ **PDF** stands for **Portable Document Format**.
- ▶ It is a **fixed-layout** format — each item is placed at a fixed position on the page.
- ▶ It keeps fonts, images, and layout together, so the file looks the **same on every device and printer**.
- ▶ It is widely used for e-books, forms, official letters, admit cards, and result sheets.

Why PDF is used

A PDF is meant to look identical everywhere, so the sender and the receiver see exactly the same page.

Why PDF Looks the Same Everywhere

Fixed layout

- ▶ Positions, spacing, and page breaks are stored in the file.
- ▶ The reader app just draws the page as instructed.

Re-flowable formats

- ▶ In a word processor, text re-flows to fit each screen.
- ▶ Layout can shift when fonts or devices change.

The fixed layout is exactly why a PDF is chosen when the exact appearance matters.

Text PDF vs Scanned PDF

Point	Text PDF (editable-source)	Scanned PDF (image-only)
Page contents	Real text and vector graphics	A photograph of each page
Select / copy text	Works	Does not work
Search text	Works	Does not work
File usually	Smaller	Larger

A scanned PDF is text only to the eye; to the computer it is just a picture of text.

OCR: Making a Scan Searchable

- ▶ **OCR** stands for **Optical Character Recognition**.
- ▶ It reads the picture of the text and converts it into **machine-readable text**.
- ▶ After OCR, the text can be selected, searched, copied, and edited.
- ▶ OCR is run on scanned pages and on photographs of documents.

Worked example

A scanned admit card will not let you copy the roll number. Run OCR, and the same PDF now lets you select and search that number.

Protecting a PDF: Encryption and Passwords

- ▶ **Encryption** scrambles the file so it cannot be read without the correct key or password.
- ▶ A **password to open** (user password) is required before the file will open at all.
- ▶ A **permissions password** (owner password) restricts printing, copying, or editing.
- ▶ Encryption provides **confidentiality** — only the authorised reader can see the content.

Trap alert

Password protection is an access control on the file. It is **not** the same thing as a digital signature.

Digital Signature: Authenticity and Integrity

- ▶ A **digital signature** is attached to a file to prove **who signed it**.
- ▶ It also proves **integrity**: if the file is changed after signing, the signature no longer matches.
- ▶ It uses a digital certificate from a trusted authority.
- ▶ It provides **authenticity** and **integrity** — it does not hide the contents.

One line to remember

A signature says *who* sent it and that it was *not altered*; it does not make the file unreadable to others.

Encryption vs Digital Signature

Point	Encryption	Digital signature
Main purpose	Confidentiality	Authenticity and integrity
Result	Contents unreadable	Signer verified, tampering detected
Needs	Key / password	Digital certificate

Trap alert (TRM-11, TRAP-15)

A digital signature does **not** encrypt the file, and encryption does **not** prove who sent it. Do not treat the two as the same.

Redaction: Removing Sensitive Content

- ▶ **Redaction** means **permanently removing** sensitive content before a file is shared.
- ▶ Typical items: names, account numbers, addresses, or photographs.
- ▶ A proper redaction deletes the underlying text or image, not just draws a black box over it.
- ▶ Covering text with a rectangle is not true redaction; the hidden text can often still be copied.

Trap alert

A black mark on the page is not safe if the text underneath can still be selected and copied.

PDF/A: Archival PDF

- ▶ **PDF/A** is a special **ISO standard** version of PDF made for **long-term archiving**.
- ▶ It embeds the fonts and colour information inside the file.
- ▶ The goal is that the document will render **exactly the same** years later, even on future software.
- ▶ PDF/A is used by libraries, courts, and government records.

Remember the “A”

The “A” in PDF/A stands for **archival** — built for permanent preservation, not everyday editing.

Format Identification Drill

Extension	Format	Category
.pdf	Portable Document Format	Fixed-layout document
.docx	Word document	Editable document
.csv	Comma-Separated Values	Plain-text data
.png	Portable Network Graphics	Image (lossless)
.jpg	JPEG image	Image (lossy, photos)

Identify the file by both the extension and the format it stands for, not by the extension alone.

Trap Alert: Facts to Keep Straight

- ▶ The **extension** is the suffix; the **format** is the data structure. (TRAP-34)
- ▶ A **scanned PDF** is an image; its text cannot be selected or searched until **OCR** is run.
- ▶ **Encryption** gives confidentiality; a **digital signature** gives authenticity and integrity. (TRM-11)
- ▶ **Redaction** removes sensitive content permanently, not just visually.
- ▶ **PDF/A** is for long-term **archiving**.

Lesson 33: Recall

- ▶ The extension is the file-name suffix; the format is the real structure of the file.
- ▶ Common document formats: DOCX (Word), XLSX (Excel), PPTX (PowerPoint), CSV, and TXT.
- ▶ PDF is a fixed-layout format and looks the same on every device.
- ▶ A text PDF is searchable; a scanned PDF is only an image until OCR converts it into readable text.
- ▶ Encryption gives confidentiality; a digital signature gives authenticity and integrity — they are not the same.
- ▶ Redaction permanently removes sensitive content; PDF/A is built for long-term archiving.